

FILTER DESIGN TEST CASE

FBAR (Film Bulk Acoustic Resonator) - Band 2 (1900 MHz) for Satellite L-Band Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a FBAR (Film Bulk Acoustic Resonator) filter targeting Band 2 (1900 MHz) for Satellite L-Band Filter. The filter is designed on Langasite substrate with a target center frequency of 5475.7 MHz and bandwidth of 240.1 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Satellite L-Band Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: FBAR (Film Bulk Acoustic Resonator)
- Frequency Band: Band 2 (1900 MHz)
- Application: Satellite L-Band Filter
- Substrate: Langasite
- Package: SiP (System in Package)
- Design Tool: Matlab RF Toolbox R2024b
- Design Center: Osaka Design Center

This specification applies to all variants of the FBAR (Film Bulk Acoustic Resonator) design targeting Band 2 (1900 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	5475.7 MHz
Bandwidth (3 dB):	240.1 MHz
Insertion Loss Target:	≤ 2.6 dB
Return Loss Target:	≥ 19.0 dB
Out-of-Band Rejection:	≥ 39.0 dB
Isolation (Duplexer):	≥ 16.0 dB
Q Factor:	12634
Temperature Coefficient:	-22.9 ppm/°C
Die Size:	1.7 x 1.2 mm
Wafer Size:	8-inch
Number of Resonators:	12
Electrode Material:	Al

Passivation: Si3N4

4. TEST PROCEDURE

1. Open Matlab RF Toolbox R2024b and load the FBAR (Film Bulk Acoustic Resonator) template project.
2. Define the substrate stack: Langasite with specified layer thicknesses.
3. Set design targets: center freq 5475.7 MHz, BW 240.1 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 2.6 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (SiP (System in Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, Langasite).
10. Perform on-wafer probing using Keysight E5080B ENA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 16427 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 2.6 dB
- Return loss in passband shall be >= 19.0 dB
- Out-of-band rejection at +/- 360 MHz offset >= 39.0 dB
- Group delay variation across passband shall be <= 8 ns
- Temperature drift over -40°C to +85°C shall be <= 15.7 MHz
- Power handling shall be >= +25 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 92%

6. TEST RESULTS

Measurement Date: 2024-04-01
Devices Tested: 30
Insertion Loss (meas): 2.31 dB
Return Loss (meas): 20.8 dB
Rejection (meas): 41.8 dB
Isolation (meas): 20.8 dB
Q Factor (meas): 12634
Group Delay Variation: 3.7 ns
Power Handling: +27 dBm
Die Yield: 80.3%

Result Classification: PASS

7. OBSERVATIONS AND FINDINGS

- a) The FBAR (Film Bulk Acoustic Resonator) design demonstrated excellent performance for Band 2 (1900 MHz).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Temperature stability was within specification across full range.
- e) Cross-talk between adjacent filter channels was below -55 dB.

8. CORRECTIVE ACTIONS

No corrective actions required. Design passed all acceptance criteria.

9. SIGN-OFF

Author:	Dr. Olaf Bergstrom	Date:	2024-04-01
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Approver:	Dr. Hans Weber	Date:	2024-04-14

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